

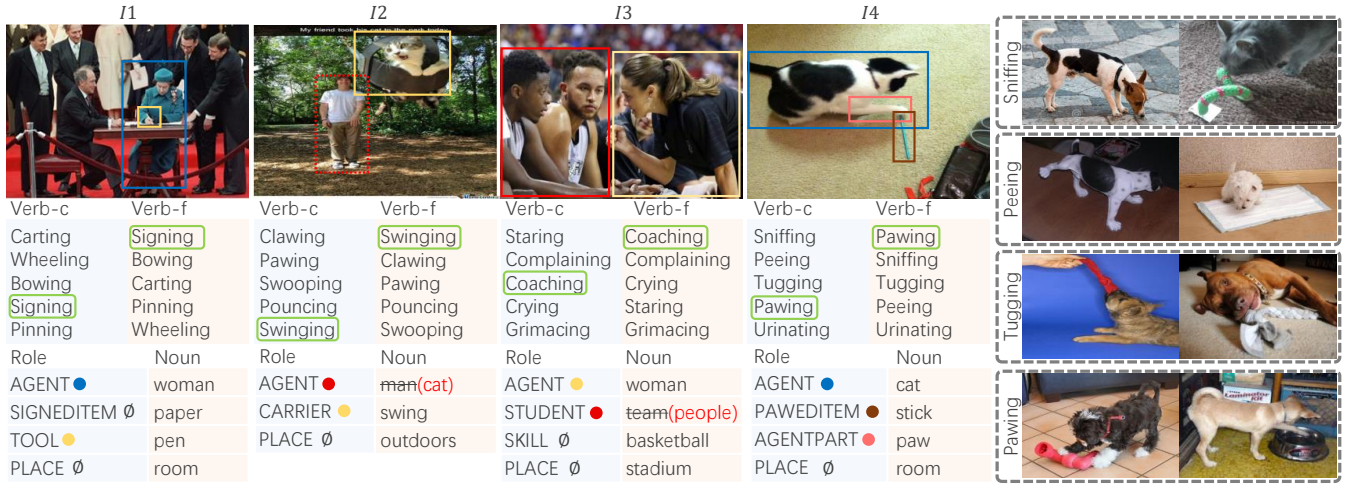


Figure 6: **Left:** For each image, the top-5 verb candidates and re-ranked verbs are shown below “Verb-c” and “Verb-f”, respectively. The semantic role detection are shown in the third row. Incorrectly object category prediction is scratched out with ground-truth shown in red brackets. The correct groundings are shown in solid boxes while the incorrect ones shown in dotted boxes. $\emptyset$ means no ground-truth grounding for that role. **Right:** The retrieved support images of top-4 verb candidates for *I4*.

# Layers	Top-1	Top-5	# Layers	Top-1	Top-5
0(JSL)	39.94	67.60	4	43.08	71.21
1	41.88	70.10	5	42.74	71.22
2	42.52	70.95	6	42.34	70.56
3	42.79	70.93			

(a) Top-1 & top-5 verb accuracy w.r.t. different layer numbers of the encoder in Verb-c (*i.e.*, w/o Verb-f).

# Support Imgs	Top-1
1	43.50
3	43.96
5	44.19
10	44.20

(b) Verb accuracy w.r.t. sizes of support images.

Table 4: Results (%) on different hyper-parameters choices of CFVM.

V-query	Shared R-query	value	val-all	grnd	grnd-all
✓	✓	75.85	42.13	61.89	24.89
✗	✓	74.17	39.37	60.16	22.86
✓	✗	73.26	38.13	57.02	20.42
✗	✗	70.96	34.87	55.37	19.02

Table 5: Results (%) of different query designs in TNM under Ground-Truth-Verb setting.

(*i.e.*, each verb has independent semantic role queries). Results under Ground-Truth-Verb setting are reported in Table 5. From the results, we can observe that the improvement of sharing role queries is obvious (*e.g.*, $\approx 2\%$ and 3% performance gains for the *value* and *val-all* metrics).

Effectiveness of parallel decoding. As shown in Table 1 & Table 2 Ground-Truth-Verb setting where the effect of verb model is exempted, the quantitative improvements to ISL and JSL attest to the effectiveness of parallel decoding.

Qualitative Results. In Figure 6, we display coarse-to-fine verb predictions and semantic role detection results of some images (*I1* ~ *I4*) from the test set (left) and retrieved support image sets (right). For all four examples, their initial top-1 verb predictions from Verb-c are wrong but the ground-truth verbs are probabilities ascend to the 1st place after re-ranking by Verb-f. We can see that discriminative details are needed to distinguish ground-truth verb from these candidates (*e.g.*, tiny interaction between the cat and the stick in *I4*). We also display some errors of TNM. In *I2*, the AGENT of Swinging is incorrectly predicted as man. This error may be caused by the rare occurrence of “a cat is swinging”. While in *I3*, the incorrect team for STUDENT is actually more reasonable than the ground-truth people. On the right part of Figure 6, we show retrieved support images of Pawing and the top-3 wrong predicted verbs.

Conclusions

In this paper, we argue that the existing two-stage GSR models have drawbacks in both verb prediction stage (insufficient to handle high diversity of daily activities) and semantic role detection stage (using autoregressive model). To alleviate these drawbacks, we propose SituFormer which consists of a two-step verb model and a transformer-based noun model. Specifically, the verb model predicts verb in a coarse-to-fine process, and the noun model makes use of the flexibility of transformer to integrate the recognition and grounding of roles. We achieved significant gains over all metrics on challenging benchmark *SWiG*, and conducted ablative analysis for each component of SituFormer.

Acknowledgements

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